

BACKGROUND

- Cancer treatments can vary in their likelihood of affecting the future reproductive abilities of patients. Approximately 40% to 80% of cancer patients that are assigned female at birth face possible infertility as a result of their cancer treatments, such as chemotherapy, radiation, and surgery.¹ An additional 15% to 30% of cancer survivors that are assigned male at birth also face infertility as a result of chemotherapy.² Therefore, it is imperative for healthcare providers that patients are made aware of fertility preservation, also coined oncofertility, techniques available at the time of their cancer diagnosis.
- While not all medications used in cancer treatment are considered gonadotoxic, some treatments do have the potential to cause long-term infertility through damage to the ovaries or testes. There are different options to preserve fertility. One option is to bank healthy cells prior to starting cancer therapy. Postpubescent patients can opt for embryo or gamete (sperm or oocyte) cryopreservation. Prepubescent patients may use tissue cryopreservation of either ovarian or testicular tissue.
- Both the American Society of Clinical Oncology (ASCO) and the National Comprehensive Cancer Network (NCCN) recommend that patients diagnosed with cancer be offered fertility preservation services before beginning their oncologic treatment.
- The Walgreens Specialty Pharmacy Connected Care Oncology Fertility Preservation intervention was designed to ensure that cancer patients had the option of consulting with a reproductive endocrinologist prior to starting their gonadotoxic treatment. The intervention did not seek to delay treatment, but rather to ensure that patients diagnosed with cancer are given a choice in their reproductive future.

OBJECTIVES

- This study aimed to understand the utilization of the Connected Care Oncology Fertility Preservation intervention, including its value in creating awareness around fertility preservation in newly diagnosed cancer patients. The patients must be new to therapy, receiving a gonadotoxic treatment, and be younger than 50 years old.
- The results will also be used to assess the intervention's impact on patient outcomes and satisfaction in the future. This includes measuring how well the intervention improves patients' understanding of their fertility preservation options, their engagement in reproductive risk counseling, and their overall satisfaction with the care they receive.
- Additionally, evaluating the intervention can help identify areas for improvement and ensure that it meets the needs of diverse patient populations.

ACKNOWLEDGMENTS

- The authors would like to thank Amy Pfeifer, PharmD, BCPS, CSP, and Ashley McClure Wolfson, PharmD for their contributions to this project.

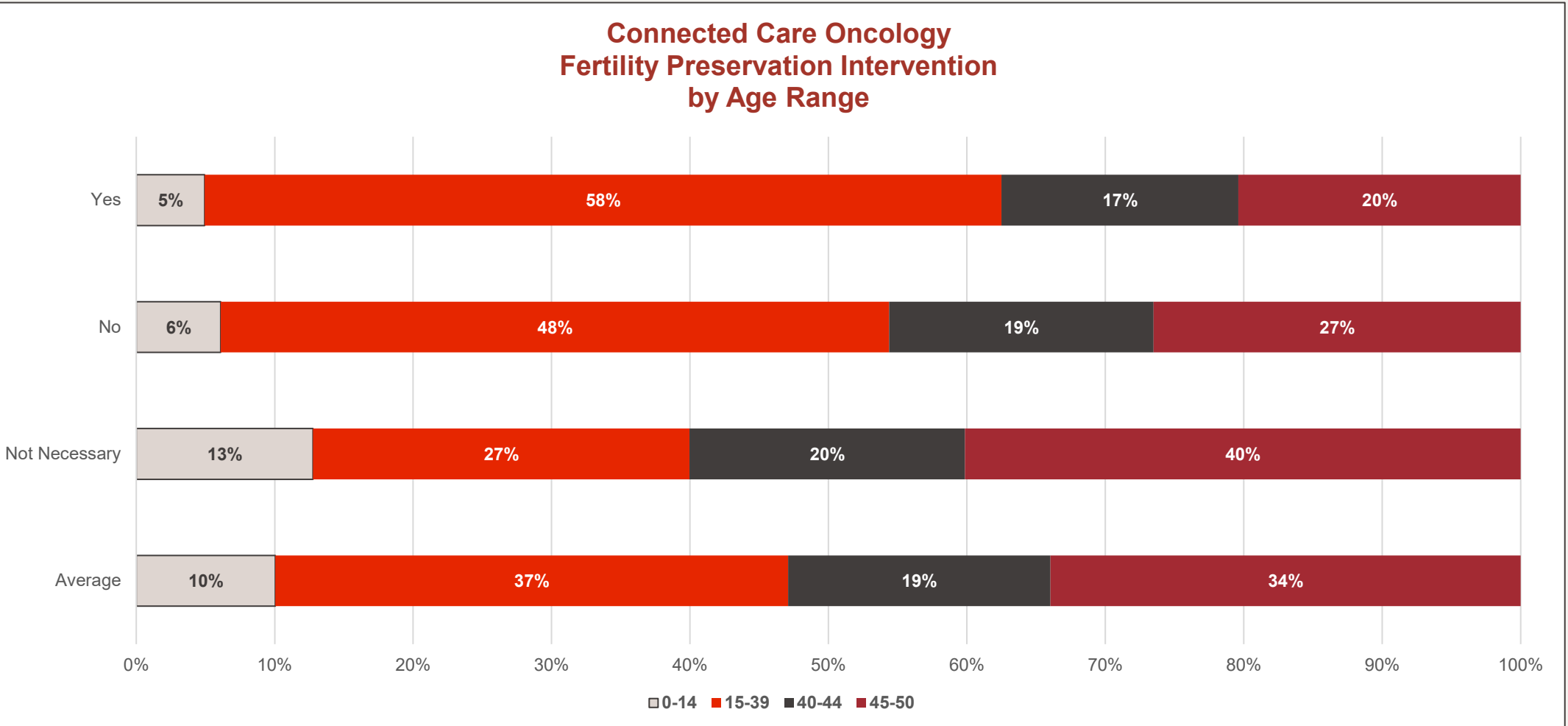
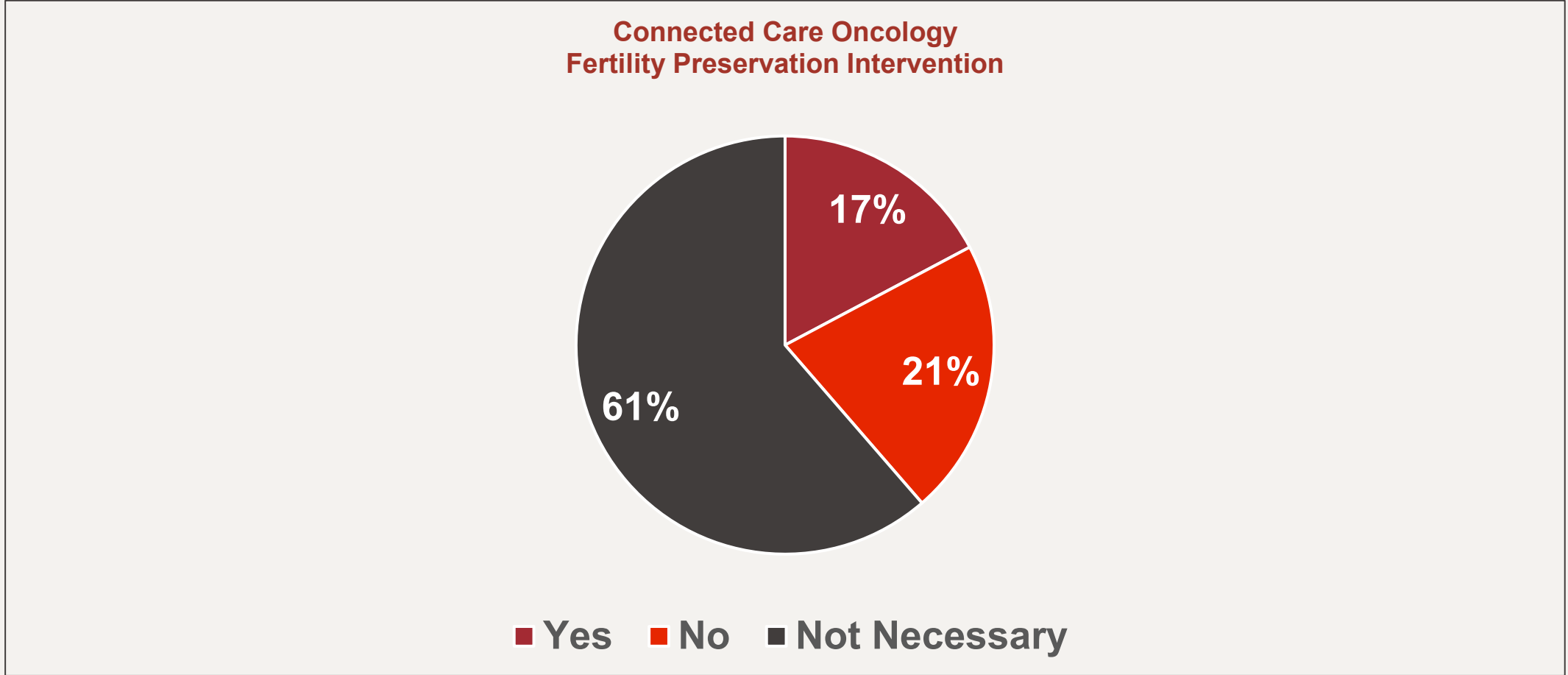
METHODS

- Adhering to medication treatment regimens is crucial for chronic disease patients to ensure proper management of their disease.
- Medication adherence rates are found to be low among patients with chronic diseases. Medication non-adherence may be a result of forgetfulness.
- Some retail community pharmacies have the technological infrastructure to send their patients automated email reminders to pick up a new or refilled prescription.
- While existing studies have examined patient receptiveness to receiving health-related information via email and the effect of these emails on behavior, little is known about the effectiveness of email refill reminders on patient medication adherence.
- This study design was descriptive, nonexperimental, and retrospective.
- Records from patients who received gonadotoxic oncologic treatment from Walgreens Specialty Pharmacy from December 2023 to May 2025 were analyzed to determine the percentage of patients who opted for a fertility preservation counseling session with a fertility specialist.
- Intervention:
 - Reproductive endocrinologists are doctors who have specialized training in helping people become and stay pregnant. They can assist with both fertility preservation and infertility treatment. Fertility preservation is the process of saving or protecting eggs, sperm, or reproductive tissue so that a person can use them to have biological children in the future. The treatment that you are receiving can affect your future fertility. Have you had an appointment to discuss this with a reproductive endocrinologist?
- Responses:
 - Yes
 - No
 - Not Necessary
- If response is NO:
 - If you are interested in fertility preservation, you should schedule a consult with a reproductive endocrinologist as soon as possible. This timing is important because the discussion needs to take place before treatment begins. Some fertility clinics offer same or next day appointments that can be conducted virtually. One option is available in the Walgreens App. To access, click on "Health Services" at the bottom of the main screen. Within the Categories section, select "Cancer Care" then choose the option with fertility services.

RESULTS

- A total number of 1,763 patients were classified as new to therapy cancer patients during this period and were asked about a fertility preservation conversation.
- Of these, 1,082 patients, or 61%, felt a conversation was not necessary. Additionally, 377 patients, or 21%, stated their oncologist had not discussed it with them, and 304 patients, or 17%, stated they had a consultation with a fertility specialist.

RESULTS (continued)



CONCLUSIONS

- This study supports already published data indicating that opportunities still exist to counsel newly diagnosed or new to therapy oncology patients on their fertility preservation options.
- Additionally, the intervention shows that pharmacists are another resource patients could look to as a solution to a general lack of awareness among oncology clinicians and patients. Pharmacists and other healthcare providers can help cancer patients by identifying those who are eligible for fertility preservation methods as well as managing medication side effects. Moreover, pharmacy team members can also educate patients on financial assistance resources, which has also been reported as a major barrier to fertility preservation.
- Ultimately, this gives cancer patients the potential opportunity to have biological children, which might not have been possible otherwise. It is also important to note that cancer survivors have additional ways of expanding their families including but not limited to egg donation, embryo adoption, and adoption.